

### LEARNING OBJECTIVES.

- *Determine the posterior probability distribution in the case of conjugate priors.*

### Motivating Example

Suppose we are interested in  $\mu$ , the true average height of female athletes at Centre. We have data from our class sample, but we want to also incorporate outside knowledge about athletes and college students in general. Following our typical Bayesian approach, we need to determine a prior distribution and a likelihood, which we can then combine into a posterior.

### The Normal Distribution

**In R:** The `plot.normal(mean,sd)` function in the `bayesrules` package plots the Normal density. The `pnorm(q,mean,sd)` function gives the probability  $P(Y \leq q)$  for  $Y \sim N(\text{mean}, \text{sd})$ .

The Normal model often is a good approximation for biological quantities such as heights from a single population. Suppose we have  $n$  students in our class data survey that are female athletes. We will assume that these  $Y_1, \dots, Y_n$  values are independent observations from  $Y \sim N(\mu, \sigma^2)$ . We will let  $\mu$  be an unknown variable, but we will assume a fixed standard deviation. ChatGPT estimated this standard deviation to be 2.8 inches, but for now we will just write  $\sigma$ . (If the standard deviation is not fixed, we would need a joint prior distribution on  $\mu$  and  $\sigma$ , which is beyond the scope of this lesson).

Give the joint pdf of these students. Then turn that into a likelihood up to a proportionality constant, simplified as much as possible.

We also need a prior distribution for the unknown  $\mu$ . In general, since  $\mu$  can take on any real number, it makes sense to use a prior for  $\mu$  whose support is  $(-\infty, \infty)$ . So we will use the Normal distribution, with mean  $\theta$  and standard deviation  $\tau$ ,

We can tune this prior based on any relevant information we have. A Google search reveals considerable differences based on particular sports, and most of the available data is on Division I athletes, so we may not want too informative of a prior. But we can be fairly confident the true average height is between 64 inches (the average height of females in general) and 72 inches (the average height of players in the WNBA). Give a reasonable prior based on this information or your own intuition.

### **The Normal-Normal Bayesian Model**

**In R:** You can use the `plot_normal_normal()` and `summarize_normal_normal()` functions in the `bayesrules` package in the same way as the previous conjugate families.

**Example 1.** Combine the prior and likelihoods for our athlete example to calculate a posterior distribution for the mean height of Centre female athletes.

**Example 2.** Again, for this example we will consider our class survey to be representative of all Centre students (which allows us to assume independence). We are interested in the average high school GPA of students at Centre. You may assume a standard deviation of 0.35 for this variable of interest.

- (a) Tune a reasonable Normal prior for the data, using intuition, an Internet search, or generative AI.
- (b) Plot a histogram or density curve of the class data to see if a Normal likelihood is reasonable. Then give the likelihood up to proportionality.
- (c) Derive the posterior distribution.
- (d) What is the posterior probability that the average high school GPA of Centre students is less than 4.0? Hint: Use `pnorm`.
- (e) Plot the relevant distributions using `plot_normal_normal`. Write a couple of sentences detailing how much your prior changed, and the relative information given by prior and likelihood.

## Deriving the Normal-Normal Model